Assignment 2 : Application case study: MedConnect, a telehealth

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1. Introduction to Prototyping

A prototype is an early, working representation of a software product that allows designers, developers, and stakeholders to evaluate an idea before committing to full scale development. Prototyping sits at the centre of modern humancomputer interaction practice because it converts abstract requirements into something a user can see, click through, and react to. Low fidelity prototypes (sketches and wireframes) validate structure and flow cheaply, while high fidelity prototypes simulate visual design, motion, and interaction closely enough to be mistaken for the finished product. Across both levels, the same goal applies: surface usability problems and misunderstood requirements while they are still inexpensive to fix, rather than after the application has been built.

This report researches three established prototyping tools, compares them across the criteria required by the assignment, and then applies one of them a custom high fidelity, codebased simulation of Figma class interactivity to design MedConnect, a healthcare application that connects patients with doctors for online appointment booking and live video consultations.

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2. Comparison of Prototyping Tool

2.1 Figma

Figma is a cloud based, collaborative interface design platform that has become the de facto industry standard for product teams. It runs entirely in the browser, which removes installation barriers and lets distributed teams coedit a single file in real time.

- Ease of use: moderate. The canvas and component model are approachable, but mastering auto-layout, variants, and interactive components takes practice.
- Main features: realtime multi user editing, reusable component libraries, auto-layout responsive frames, smart-animate transitions, and a developer handoff inspector that exports CSS and design tokens.
- Advantages: works on any operating system through the browser, scales well for large design systems, and integrates directly with engineering workflows.
- Limitations: requires a stable internet connection for full functionality, and very large files can slow down on lower end hardware.

2.2 Adobe XD

Adobe XD is a desktop first UX/UI design and prototyping application built for vector based screen design and click-through prototypes, with strong ties to the rest of the Adobe Creative Cloud suite.

- Ease of use: high, particularly for users already familiar with Photoshop or Illustrator conventions.
- Main features: repeat grids for fast list and card layouts, voice-triggered prototyping, auto animate transitions, and reusable component states.
- Advantages: smooth offline performance, tight integration with Adobe's asset and font ecosystem, and a lower learning curve for layout work.
- Limitations: Adobe has slowed active development of XD, so its collaborative and plugin ecosystem now trails Figma's pace of new features.

2.3 Balsamiq

Balsamiq is a deliberately low fidelity wireframing tool that mimics hand sketched interfaces, keeping teams focused on structure and flow rather than visual polish during early discovery.

- Ease of use: very high. A large library of pre built, sketch style UI components can be dragged onto the canvas with no design background required.
- Main features: a hand drawn visual skin, click through linking between mock ups, and a component library covering most common UI patterns.
- Advantages: it is fast for brainstorming and keeps early stage feedback focused on usability and content rather than colour or branding.
- Limitations: it cannot produce realistic, high fidelity, pixel accurate mock ups, so teams must switch tools once visual design begins.

2.4 Side-by-side Summary

Criterion	Figma	Adobe XD	Balsamiq
Ease of use	Moderate	High	Very high
Fidelity	High	High	Low
Collaboration	Real time, cloud based	Limited, file sharing	Limited, file sharing
Best for	Production ready UI design	Offline visual design	Early structure and flow
Main limitation	Requires internet connection	Slower feature development	Cannot show high fidelity visuals

3. Conclusion of Tool Research

Each tool serves a different stage of the design process. Balsamiq is the fastest route to validating structure and flow before any visual decisions are made. Adobe XD offers a comfortable, offline friendly middle ground for teams already inside the Adobe ecosystem. Figma was selected as the primary direction for this assignment because its component system, realtime collaboration, and browser based accessibility make it the closest match to how production interface teams actually work today, and because its interaction model maps cleanly onto the code based, click through simulation used to build the MedConnect prototype below.

4. Prototype Design: MedConnect

MedConnect is an imaginary mobile healthcare application that lets a patient search for a doctor by specialty, review their profile and rating, book an appointment in an available time slot, and join a live video consultation, all from a single phone sized interface. The prototype below was built as a high fidelity, fully interactive simulation, the same fidelity level a finished Figma or Adobe XD click through prototype would deliver, so it can be demonstrated directly inside the presentation video.

4.1 Application Flow

- Screen 1 : Welcome & Login: the patient enters their email and password and taps Log in.
- Screen 2 : Home Dashboard: a welcome banner greets the patient by name; specialty categories (General Medicine, Pediatrics, Cardiology) and a list of available doctors are presented; tapping a doctor card opens their profile.
- Screen 3 : Doctor Profile & Booking: the patient reviews the doctor's photo, rating, and experience, selects a date and time slot from an interactive calendar, and taps Book appointment.
- Screen 4 : Live Video Consultation: a full screen video call view shows the doctor, a picture in picture box for the patient, mute and camera toggle controls, and an End call button that returns to the dashboard.

4.2 Visual Design Direction

The interface uses a teal and deep forest green palette to evoke trust, calm, and clinical professionalism, paired with a warm serif display face for headlines and a clean grotesque sans-serif for body text and UI labels, a pairing intended to feel closer to a premium consumer health brand than a generic admin panel. The screens are framed inside a realistic mobile device shell so the flow can be screen recorded exactly as it would appear on a real phone.